Failure-Aware and Confidence-Calibrated Adaptive Retrieval-Augmented Generation System

23CSE399 Project Phase 1 – Approval Presentation

Team ID: 23UG031 **Panel No.:20**

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Screenshot of your guides' approval mail (to go ahead with the presentation)

Problem Definition

Retrieval-Augmented Generation (RAG) enhances the quality of responses by retrieving relevant documents from external sources based on the similarity of embeddings.

However, the current RAG models rely only on the similarity scores for ranking documents and do not evaluate the reliability of the retrieved results or identify retrieval failure.

This leads to the possibility of substandard, inconsistent, or irrelevant results being retrieved unnoticed, resulting in hallucinations and unreliable responses.

There is a need for a framework for retrieval failure detection that represents retrieval failure signals such as similarity score, consistency of top-k similarity, and retrieval agreement.

This paper proposes the design of a retrieval failure signal estimation module to calculate retrieval reliability and retrieval failure.

Literature Survey

S.No.	Authors name(s)	Full title of the paper with year	Inference from the paper	Open Problem (For proposed work)
1	Hadi Soudani, François Yvon, et al.	*Why Uncertainty Estimation Methods Fall Short in Retrieval-Augmented Generation*, 2025	Traditional uncertainty estimation (UE) fails in RAG systems because it ignores non-parametric retrieved knowledge. The paper defines axiomatic constraints for reliable UE and shows that most existing methods violate them.	Current UE methods rely only on model outputs and ignore retrieval quality, evidence consistency, and grounding. A retrieval-aware, calibrated uncertainty framework is needed for reliable RAG systems.
2	Leandre Ngartera, Ahmed Abdelaziz, et al.	*Bayesian RAG: Uncertainty-Aware Retrieval for Reliable Financial Question Answering*, 2026	Bayesian RAG models retrieval uncertainty using Monte Carlo Dropout and probabilistic embeddings, improving calibration, and reducing hallucinations.	It ignores generation instability and adaptive control. A unified uncertainty framework is needed.

Literature Survey

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3	Michael Oehri, Jannis Vamvas, et al.	*Trusted Uncertainty in Large Language Models: A Unified Framework for Confidence Calibration and Risk-Controlled Refusal*, 2025	This paper proposes a unified framework combining sequence likelihood, semantic entropy, and self-consistency to produce calibrated confidence scores, along with risk-controlled refusal mechanisms..	it does not incorporate retrieval-aware uncertainty or adaptive retrieval. A RAG system that dynamically adjusts retrieval, generation, and responses based on confidence is needed.
4	Lewis, P., Perez, E., Piktus, A., et al.	Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks, 2020	Introduces the RAG framework that combines retrieval and generation to improve factual accuracy and reduce hallucinations in knowledge-intensive tasks.	Does not provide calibrated confidence estimation or retrieval-aware uncertainty handling. Adaptive response control based on confidence is still needed.

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Justification for the Proposed Problem

The current RAG models employ similarity scores for ranking but lack the ability to identify retrieval failure explicitly.

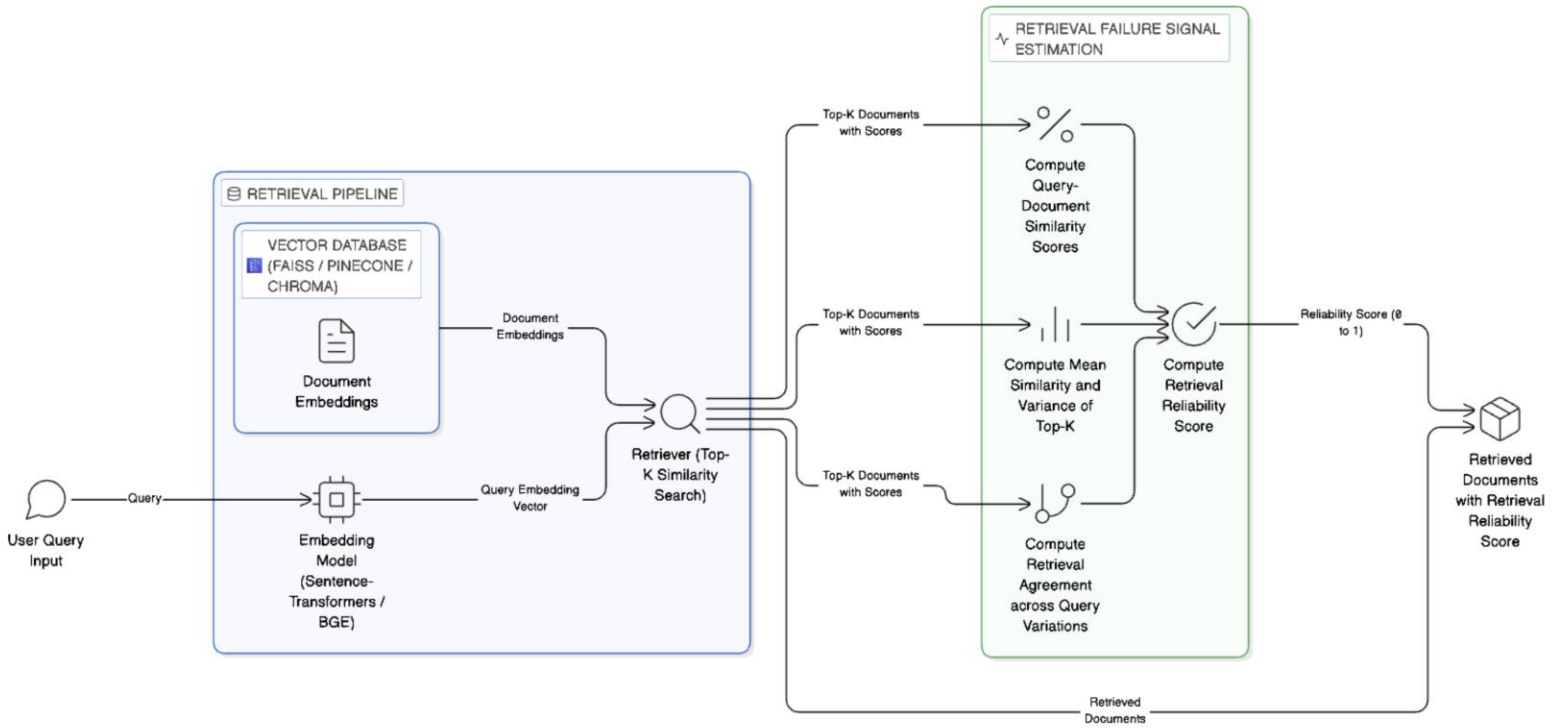
The recent models like Self-RAG and CRAG assess the quality of retrieval but lack the ability to compute retrieval reliability explicitly from measurable signals.

This results in the potential of poor retrieval going unnoticed, thereby causing unreliable responses.

There is a requirement for a system that computes the retrieval failure signal based on similarity score, top-k consistency, and retrieval agreement.

The plan for the implementation of the estimation of the retrieval failure signal for the identification of retrieval failures is proposed in this paper.

Architecture of the Proposed System



Functionalities of the Proposed System

System capabilities:

- Retrieve documents using embedding similarity
- Compute query-document similarity scores
- Analyze top-k similarity consistency
- Compute retrieval agreement signals
- Estimate retrieval reliability score
- Detect weak or unreliable retrieval
- Enable retrieval failure detection

Software/Tools Requirements

Programming Language:

- Python

Frameworks:

- HuggingFace Transformers
- Sentence-Transformers
- LangChain / LlamaIndex
- FAISS / ChromaDB

Models:

- Embedding Model: BGE-large or MiniLM
- Generator Model: Mistral-7B (future integration)

Platform:

- Kaggle GPU Environment

Data Set

Source of data set:

URL:

https://nlp.cs.washington.edu/triviaqa/

Description:

TriviaQA is a reading comprehension dataset containing over 650K question-answer-evidence triples. TriviaQA includes 95K question-answer pairs authored by trivia enthusiasts and independently gathered evidence documents, six per question on average, that provide high quality distant supervision for answering the questions.

Timeline

- Abstract Submission • 5/2/2026
- Literature Survey • 17/2/2026
- System Architecture Design Finalization • 5/3/2026
- Embedding Model Integration and Vector Database Setup • 20/3/2026
- Basic Retrieval Pipeline Implementation • 5/4/2026
- Implementation of Retrieval Similarity Score and Top-K Similarity Consistency Signal • 15/4/2026
- Implementation of Retrieval Agreement Signal • 30/4/2026

Conclusion

This work proposes a failure-aware RAG framework to improve system reliability.

This phase focuses on detecting retrieval failures using similarity score, top-k consistency, and retrieval agreement signals.

The system estimates retrieval reliability and enables early detection of weak retrieval.

This forms the foundation for future confidence estimation and adaptive RAG systems.

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THANK YOU